Software Requirements Specification

for

MomentsHub: A Collaborative Daily Photo Capsule

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

*This Software Requirements Specification (SRS) document captures the complete functional and non-functional requirements for **MomentsHub: A Collaborative Daily Photo Capsule**, Version 1.0. This document describes the intended purpose, features, and interfaces of the product, providing a foundational guide for the project's design, development, and testing phases.*

1.2 Document Conventions

*This document follows the IEEE Std 830-1998 standard for SRS documents. Functional requirements are uniquely identified with the prefix **FR-X**. All text uses the Times New Roman font.*

1.3 Intended Audience and Reading Suggestions

This document is intended for the following roles within this academic project:

- ***Development Team (2 Members):** As the developers, you will use this document to understand what needs to be built (Sections 3, 4, 5) and to manage the project scope (all sections). In this two-person team, you share the roles of Project Manager, Developer, and Tester.*
- ***Faculty/Project Guide:** As the primary stakeholder, your faculty will use this document to understand the project's vision, scope, and planned features, and to evaluate the final product against these requirements (Sections 1, 2, 4).*

1.4 Product Scope

MomentsHub is a browser-based web application designed to provide a private, intimate space for small groups of friends and family to share one photo per day. The product aims to foster genuine connection by creating a collaborative daily journal of life's small moments. Key benefits include strengthening social bonds, creating a shared digital scrapbook, and offering a focused, ad-free alternative to mainstream social media. This project directly supports the educational goal of gaining practical, end-to-end experience in modern full-stack web development.

1.5 References

IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

2. Overall Description

2.1 Product Perspective

MomentsHub is a new, self-contained product. It will be developed as a modern full-stack web application, with a React frontend and a serverless Firebase backend. It is designed to be a lightweight application that leverages cloud services for its core functionality.

2.2 Product Functions

The major functions of MomentsHub include:

- *Secure user registration and authentication.*
- *Creation of and joining private, invite-only groups.*
- *Daily photo upload with captions and tags.*
- *A rich interaction system including comments, emoji reactions, and user mentions.*
- *A gallery with advanced search and filtering capabilities.*
- *Features to promote nostalgia, such as an "On This Day" view.*
- *Gamification elements like streaks and achievements to encourage daily participation.*

2.3 User Classes and Characteristics

User Class	Characteristics
Group Member	*General user. Can join groups, upload photos, comment, react, and view all content within their groups. Assumed to be tech-savvy enough to use modern web applications but not necessarily a technical expert.*
Group Admin	*The user who creates a group. Has all the abilities of a Group Member, plus the ability to manage the group (e.g., remove members, regenerate invite code).*

2.4 Operating Environment

- ***Platform:*** *Web Browser (Desktop and Mobile)*
- ***Supported Browsers:*** *Latest versions of Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.*
- ***Backend:*** *The application will run on Google's Firebase cloud infrastructure. No specific*

server-side operating system management is required.

2.5 Design and Implementation Constraints

- *The application must be developed using React.js for the frontend.*
- *The backend services (Auth, Database, Storage) must be implemented using Google Firebase.*
- *The project must be developed at zero cost, relying solely on the free tiers of all chosen services.*
- *The application must be responsive and fully functional on both mobile and desktop screen sizes.*
- *All data transfer must be secured using HTTPS.*

2.6 User Documentation

A README.md file will be provided in the project's code repository. This document will serve as a user and developer guide, explaining how to set up, run, and use the application.

2.7 Assumptions and Dependencies

- *Users are assumed to have a stable internet connection to access the application.*
- *The project is dependent on the continued availability and free-tier limits of Google Firebase services.*
- *The development team is assumed to have proficiency in JavaScript, React, and basic database concepts.*

3. External Interface Requirements

3.1 User Interfaces

The application will feature a clean, modern, and intuitive user interface built with Tailwind CSS. The UI will be responsive and adhere to a consistent design language. Key screens include:

- *Login/Signup Page*
- *Dashboard/Home Screen (featuring "On This Day" and a feed)*
- *Group Gallery View (with search and filtering)*

- *Photo Detail View (with comments and reactions)*
- *User Profile & Settings Page*

3.2 Hardware Interfaces

There are no direct hardware interfaces beyond standard web browser interactions with keyboards, mice, and touch screens.

3.3 Software Interfaces

- ***Firebase Authentication:*** *Used for user identity management (email/password).*
- ***Firebase Firestore:*** *The NoSQL database used for storing all application data (user profiles, group info, photo metadata, comments, etc.).*
- ***Firebase Storage:*** *Used for storing the uploaded photo files.*

3.4 Communications Interfaces

- *All communication between the client (browser) and the Firebase backend will be conducted over HTTPS.*
- *Real-time data synchronization for features like notifications and comments will be handled by Firestore's real-time listeners.*

4. System Features

4.1 System Feature 1: User & Group Management

4.1.1 Description and Priority

High. This feature covers the fundamental ability for users to create accounts and form private groups.

4.1.2 Stimulus/Response Sequences

- ***Stimulus:*** *A new user navigates to the application and clicks "Sign Up".*
- ***Response:*** *The system displays a registration form for email and password.*
- ***Stimulus:*** *A returning user clicks "Log In" and enters their credentials.*
- ***Response:*** *The system authenticates the user and directs them to their dashboard.*
- ***Stimulus:*** *An authenticated user clicks "Create Group".*
- ***Response:*** *The system presents a form to name the group and creates it, generating a unique invite code.*

4.1.3 Functional Requirements

- ***FR1.1:*** *The system shall allow new users to register using an email and password.*
- ***FR1.2:*** *The system shall authenticate returning users via their email and password.*
- ***FR1.3:*** *Authenticated users shall be able to create a new, private group. The creator is assigned the 'Admin' role.*
- ***FR1.4:*** *The system shall generate a unique, non-guessable invite code for each group.*
- ***FR1.5:*** *User s shall be able to join an existing group by providing a valid invite code.*

4.2 System Feature 2: Core Photo Interaction System

4.2.1 Description and priority

This feature forms the core user experience of sharing and interacting with photos.

4.2.2 Stimulus / Response sequences

- ***Stimulus****: User clicks the "Upload Photo" button.*
- ***Response****: System displays a file picker. The user selects a photo.*
- ***Response****: System shows a preview of the photo and a caption input field.*
- ***Stimulus****: User fills in the caption and clicks "Post".*
- ***Response****: The photo is uploaded and appears in the group gallery. Other group members receive a notification.*
- ***Stimulus****: User long-presses a photo.*
- ***Response****: System displays a palette of emoji reactions.*
- ***Stimulus****: User types @ in a comment box.*
- ***Response****: System displays a list of group members to tag.*

4.2.3 Functional Requirements

- ***FR2.1:*** *A user shall be able to upload one JPG or PNG photo per day to a group they are a member of.*
- ***FR2.2:*** *Each photo upload must be accompanied by a text caption.*
- ***FR2.3:*** *Users shall be able to view all photos in a chronological gallery view.*
- ***FR2.4:*** *Users shall be able to add text-based comments to any photo.*
- ***FR2.5:*** *Users shall be able to react to a photo with one of five predefined emojis (e.g., ♡, 😂, 🎉, 😮, 😢). The system will display a count for each reaction.*

- ***FR2.6:*** *In a comment, a user shall be able to mention another group member by typing @ followed by their name.*
- ***FR2.7:*** *The system shall send a notification to any user who is mentioned in a comment.*

4.3 System Feature 3: Memory & Discovery System

4.3.1 Description and Priority

These features enhance long-term engagement by making it easy to rediscover past content. ***Priority: Medium.***

4.3.2 Stimulus/Response Sequences

- ***Stimulus:*** *User opens the application.*
- ***Response:*** *The system checks the current date and displays any photos from previous years on the same date in the "On This Day" section.*
- ***Stimulus:*** *User navigates to the gallery and opens the filter panel.*
- ***Response:*** *The system displays options to filter by date, user, and tags.*
- ***Stimulus:*** *User selects a date range and clicks "Apply".*
- ***Response:*** *The gallery updates to show only photos from within that date range.*

4.3.3 Functional Requirements

- ***FR3.1:*** *The main dashboard shall feature an "On This Day" section that displays photos uploaded on the same month/day in previous years.*
- ***FR3.2:*** *The gallery shall provide a search interface to filter photos.*
- ***FR3.3:*** *The search interface shall allow filtering by a date range (start/end).*
- ***FR3.4:*** *The search interface shall allow filtering by the user who uploaded the photo.*
- ***FR3.5:*** *The search interface shall allow filtering by one or more user-defined tags*

4.4 System Feature 4*:* Gamification and Engagement Engine

4.4.1 Description and Priority

These features are designed to motivate daily participation through positive reinforcement. **Priority: Medium.**

4.4.2 Stimulus/Response Sequences

- ***Stimulus:*** *A user uploads a photo for the third consecutive day.*
- ***Response:*** *The system displays a "🔥3" streak icon next to their name.*
- ***Stimulus:*** *The total number of photos in a group reaches 100.*
- ***Response:*** *The system unlocks the "Century Club" achievement badge and displays it*

on the group's page.

4.4.3 Functional Requirements

- ***FR4.1:*** *The system shall track and display a "contribution streak" (e.g., 🔥3) for users who post on consecutive days.*
- ***FR4.2:*** *The system shall automatically award achievement badges to groups upon reaching predefined milestones (e.g., "100 Photos Uploaded", "First Anniversary").*
- ***FR4.3:*** *Awarded badges shall be visible on the group's main page.*

4.5 System Feature 5: User Profile & Personalization

4.5.1 Description and Priority

*These features allow users to personalize their identity within the app**. Priority: Low.***

4.5.2 Stimulus/Response Sequences

- ***Stimulus:*** *User navigates to their profile page and clicks "Edit Profile".*
- ***Response:*** *The system displays fields to change their display name and upload a new avatar.*
- ***Stimulus:*** *User clicks on their status emoji.*
- ***Response:*** *The system displays a palette of emojis to choose from for their new status.*

4.5.3 Functional Requirements

- ***FR5.1:*** *Users shall be able to set a public display name and upload a profile avatar.*
- ***FR5.2:*** *Users shall be able to select a single emoji from a predefined list to serve as their temporary status, which is displayed next to their avatar.*

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- *The main gallery page, with thumbnails, should load in under 3 seconds on a standard broadband connection.*
- *Photo uploads should complete within 10 seconds for images up to 5MB.*
- *Real-time events like new comments should appear for other users within 2 seconds.*

5.2 Security Requirements

- *All user passwords must be securely hashed and stored by Firebase Authentication.*

- *A user must only be able to view photos and data from groups they are a member of. This will be enforced via Firestore Security Rules.*
- *A user must only be able to upload photos to groups they are a member of.*
- *All data transmission between the client and server must be encrypted via HTTPS.*

5.3 Software Quality Attributes

- ***Usability:*** *The application should be intuitive and require no formal training for a user familiar with common web applications.*
- ***Reliability:*** *The application should have an uptime of 99.5%, dependent on Firebase services.*
- ***Maintainability:*** *The code should be well-commented and structured into modular React components to facilitate future updates.*
- ***Portability:*** *As a web application, it is inherently portable across any device with a modern web browser.*

5.4 Business Rules

- *A user is limited to uploading one photo per group, per 24-hour period.*
- *The user who creates a group is designated as the 'Admin' and is the only one who can manage group members.*
- *Invite codes are permanent unless regenerated by the Group Admin.*

6. Other Requirements

- *All project development will adhere to the Iterative SDLC model.*

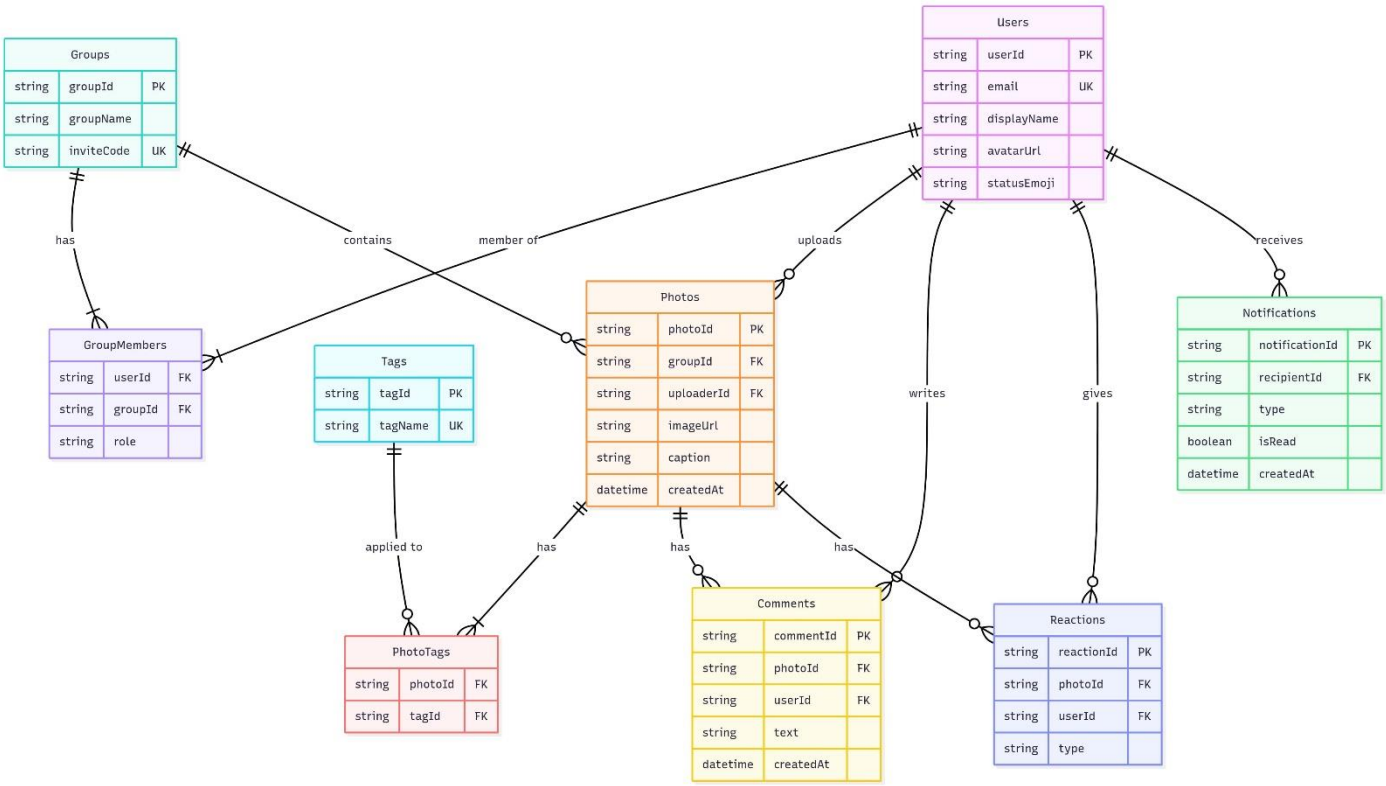
Appendix A: Glossary

- ***SRS:*** *Software Requirements Specification*
- ***UI:*** *User Interface*
- ***WBS:*** *Work Breakdown Structure*
- ***Firebase:*** *A Backend-as-a-Service (BaaS) platform by Google.*
- ***Firestore:*** *The NoSQL document database component of Firebase.*
- ***Streak:*** *A count of consecutive days a user has performed an action.*

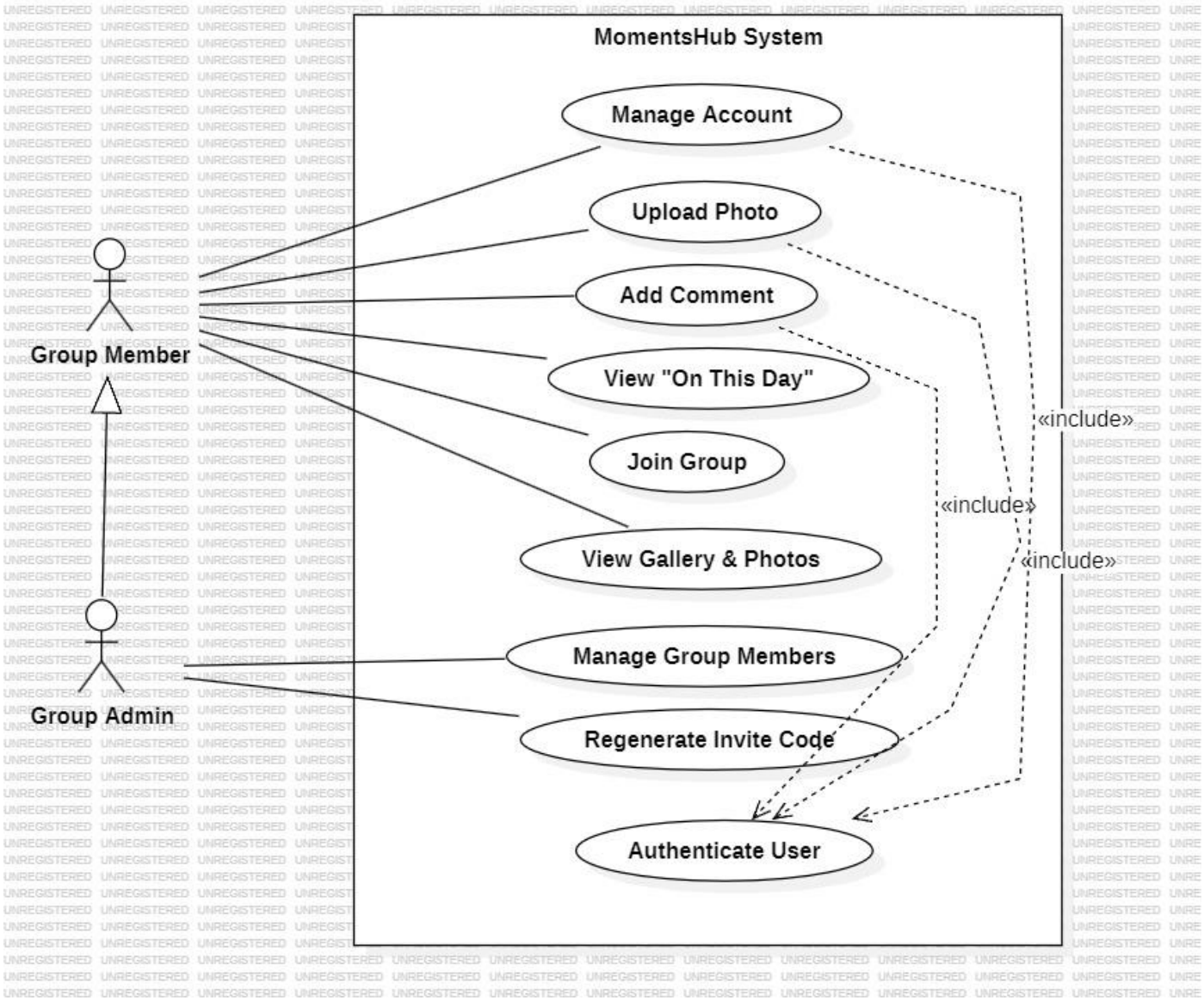
Appendix B: Analysis Models

Entity-Relationship Diagram (ERD)

The following diagram describes the database schema for the MomentsHub application.



Use Case Diagram(Star UML):-



Appendix C: To Be Determined List

1. Detailed Frontend Architecture.

2. Database & Schema Implementation.

3. Backend Logic & Cloud Functions.